Algorithm Design References — Quantum Knapsack (Grover maximum-finding)

These are the works the implementation in `knapsack.c` is built from. arXiv IDs and years are given for the reader's convenience; consult the primary sources for the exact statements relied on (the conference paper cites the core search works, items 1–4, 9–11 and 16, with pinpointed references).

A. Amplitude amplification / quantum search (the core loop)

1. **L. K. Grover**, "A fast quantum mechanical algorithm for database search," Proc. 28th ACM STOC (1996) 212–219. arXiv:quant-ph/9605043. → The amplitude-amplification primitive: oracle phase-flip + diffuser; about (pi/4)*sqrt(2^N / M) iterations concentrate amplitude on the M marked states.
2. **M. Boyer, G. Brassard, P. Høyer, A. Tapp**, "Tight bounds on quantum searching," Fortschr. Phys. 46 (1998) 493–505. arXiv:quant-ph/9605034. → The BBHT schedule for an UNKNOWN number of marked states M — the exponentially growing iteration cap we use instead of a fixed count (a fixed count assuming M=1 fails as soon as several subsets qualify).
3. **G. Brassard, P. Høyer, M. Mosca, A. Tapp**, "Quantum amplitude amplification and estimation," AMS Contemp. Math. 305 (2002) 53–74. arXiv:quant-ph/0005055. → Formal generalization of Grover to an arbitrary state-prep operator; the basis for treating the diffuser as a reflection about the prepared (uniform) state.

B. Quantum maximum-finding (the outer threshold loop)

4. **C. Dürr, P. Høyer**, "A quantum algorithm for finding the minimum," arXiv:quant-ph/9607014 (1996). → The outer loop: repeatedly run Grover marking states better than a moving threshold T, keep the best found; converges to the exact optimum in O(sqrt(2^N)) oracle calls. We adapt minimum-finding to maximum-finding (mark value >= T).

C. Reversible arithmetic inside the oracle (cost & value sums + comparison)

5. **V. Vedral, A. Barenco, A. Ekert**, "Quantum networks for elementary arithmetic operations," Phys. Rev. A 54 (1996) 147. arXiv:quant-ph/9511018. → Foundational reversible adder/accumulator networks; the controlled-addition pattern (item qubit i, when |1>, adds the constant c_i into a register).

6. **S. A. Cuccaro, T. G. Draper, S. A. Kutin, D. P. Moulton**, "A new quantum ripple-carry addition circuit," arXiv:quant-ph/0410184 (2004). → The ripple-carry adder we use (X / CX / CCX only). Its carry-OUT bit is exactly the comparison flag: add the two's-complement of (W+1) and read the high bit to test cost <= W (and likewise value >= T).
7. **T. G. Draper**, "Addition on a quantum computer," arXiv:quant-ph/0008033 (2000). → The QFT (Fourier-basis) constant-adder — the ancilla-free ALTERNATIVE. Documented for completeness; we chose ripple-carry for integer-exactness and readability.

D. Multi-controlled gate decomposition (diffuser & marking)

8. **A. Barenco, C. H. Bennett, R. Cleve, D. P. DiVincenzo, N. Margolus, P. Shor, T. Sleator, J. A. Smolin, H. Weinfurter**, "Elementary gates for quantum computation," Phys. Rev. A 52 (1995) 3457. arXiv:quant-ph/9503016. → Decomposition of multi-controlled operations (the diffuser's C^(N-1)Z and Toffoli ladders) into 1- and 2-qubit gates, for sizes where native CCX/C3X don't suffice.

E. Textbook treatment

9. **M. A. Nielsen, I. L. Chuang**, "Quantum Computation and Quantum Information," 10th Anniversary ed., Cambridge Univ. Press (2010), Chapter 6; §4.2–4.3. → Canonical Grover oracle + diffuser ("inversion about the mean") exposition (Ch. 6); the single-qubit and controlled-gate matrices the simulator applies (§4.2, §4.3).

F. Related work we deliberately did NOT implement (for honest context)

10. **S. Wilkening et al.**, "A quantum algorithm for solving 0-1 Knapsack problems," npj Quantum Information 11, 146 (2025). arXiv:2310.06623. → The Quantum Tree Generator prepares a superposition over ONLY feasible subsets — more sophisticated than our plain feasibility-oracle Grover. Intentionally not used (materially harder to validate on a from-scratch simulator).
11. **E. Farhi, J. Goldstone, S. Gutmann**, "A quantum approximate optimization algorithm," arXiv:1411.4028 (2014). → QAOA (the variational "Option B"). Not chosen: it is approximate, whereas Grover maximum-finding returns the EXACT optimum on small N.

G. Statevector simulation (the in-process backend, knapsack.c Section 1)

12. **K. De Raedt, K. Michielsen, H. De Raedt, B. Trieu, G. Arnold, M. Richter, Th. Lippert, H. Watanabe, N. Ito**, "Massively parallel quantum computer simulator," Comput. Phys. Commun. 176 (2007) 121–136. arXiv:quant-ph/0608239. → The standard description of statevector simulation by index-pair updates: a gate on qubit t pairs each amplitude with bit t = 0 against its partner with bit t = 1. Updated as **H. De Raedt et al.**, "Massively parallel quantum computer simulator, eleven years later," Comput. Phys. Commun. 237 (2019) 47–61. arXiv:1805.04708.
13. **M. Smelyanskiy, N. P. D. Sawaya, A. Aspuru-Guzik**, "qHiPSTER: The Quantum High Performance Software Testing Environment," arXiv:1601.07195 (2016). → The single-qubit and controlled-gate kernels written out explicitly (the loops the sv_* routines implement), with the stride and memory-traffic analysis.
14. **T. Jones, A. Brown, I. Bush, S. Benjamin**, "QuEST and High Performance Simulation of Quantum Computers," Sci. Rep. 9, 10736 (2019). arXiv:1802.08032. → An open-source simulator built on the same kernel; a reference implementation to compare against.
15. **S. Vigna**, "An Experimental Exploration of Marsaglia's xorshift Generators, Scrambled," ACM Trans. Math. Softw. 42(4) (2016). DOI 10.1145/2845077. arXiv:1402.6246. → The xorshift64 *generator (shifts 12/25/27, multiplier 0x2545F4914F6CDD1D) used for the reproducible random choices of the search (grng_* in knapsack.c); classical, affects which candidate is tried next, never the answer.

H. Circuit interchange (the out-of-process paths 1 and 2, knapsack.c export_solver_circuit / write_qasm)

16. **A. W. Cross, A. Javadi-Abhari, T. Alexander, N. de Beaudrap, L. S. Bishop, S. Heidel, C. A. Ryan, P. Sivarajah, J. Smolin, J. M. Gambetta, B. R. Johnson**, "OpenQASM 3: A Broader and Deeper Quantum Assembly Language," ACM Trans. Quantum Comput. 3(3) (2022). DOI 10.1145/3505636. arXiv:2104.14722. → The circuit format the solver writes for paths 1 and 2. Online: the OpenQASM Live Specification, https://openqasm.com/ (standard gate library "stdgates.inc": https://openqasm.com/language/standard_library.html); source and issues: https://github.com/openqasm/openqasm. write_qasm emits only stdgates.inc gates plus two custom definitions (ccz, mcx), so any OpenQASM 3 consumer can load the file.

Honest framing (must hold in code & paper)

- On a statevector simulator there is NO quantum speedup; exact 0/1 knapsack is NP-hard. The defensible claim is a genuine quantum ALGORITHM (amplitude amplification)

executed for validation, with the standard quadratic Grover envelope O(2^(N/2)) oracle calls.

- The oracle is BLIND to the classical optimum: it computes cost/value FROM the item qubits in superposition. The classical argmax is retained ONLY as an independent ground-truth checker, never as a seed into the circuit.